Insulin

Special note: Special care with calculating doses and administration volumes in neonates - frequently involved in medication errors.

BRAND NAME Actrapid®

FORM Vial containing insulin 1000units/10ml

INDICATION Management of Hyperglycaemia - see below.

Management of life-threatening hyperkalaemia using glucose and insulin - See monograph for Insulin and Glucose for hyperkalaemia

DOSE RANGE

AGE	DOSE	FREQUENCY	ROUTE
0 – 6months	0.02 - 0.2 units/kg/hr*	Continuous infusion	IV

Commence treatment and titrate upwards as per WOS guideline.
* Doses up to 0.5 units/kg/hr may be used on Consultant advice

PRESCRIPTION OF CONTINUOUS INFUSION

LOW CONCENTRATION (if doses <0.05 units/kg/hr are required)
10units/kg in 50ml infusion fluid

This gives:- 0.02 units/kg/hr at 0.1 ml/hour
0.1 units/kg/hr at 0.5 ml/hour
0.2 units/kg/hr at 1 ml/hour

HIGH CONCENTRATION
25units/kg in 50ml infusion fluid

This gives:- 0.05 units/kg/hr at 0.1 ml/hour
0.1 units/kg/hr at 0.2 ml/hour
0.2 units/kg/hr at 0.4 ml/hour

RECONSTITUTION Already in solution, further dilution required.

DILUTION For all IV infusions a two-step dilution is required,
Step one: Dilution of insulin 100 units/ml injection to produce a suitable solution for further dilution.
Step two: Further dilution to produce the concentrations required for continuous infusion as outlined above.

Initial Dilution:

Insulin soluble (human Actrapid) injection (100units/ml)	1ml
Water for injection	up to 10ml total

Gives 10 units in 1 ml solution. Use the required volume.

Neonatal

Final Dilution:

LOW CONCENTRATION
Using the 10 units in 1 ml (diluted) solution

1 ml/kg further diluted to 50 ml with sodium chloride 0.45% (equivalent of 10 units/kg in 50 ml)

HIGH CONCENTRATION
Using the 10 units in 1 ml (diluted) solution
2.5 ml/kg further diluted to 50 ml with sodium chloride 0.45% (equivalent to 25units/kg in 50 ml)
NB: **Prime line and leave for 10 minutes then flush the administration line through with 5ml of the solution before connecting to the patient.** Insulin adheres to plastic so this will ensure consistent IV delivery. Repeat this procedure when lines are changed.

For greater insulin requirements a more concentrated solution can be prepared – contact unit pharmacist for advice

METHOD OF ADMINISTRATION

For continuous Infusion
By continuous intravenous infusion, flow rate adjusted according to the baby's response (see prescription section for details).
DO NOT FILTER THE INFUSION

COMPATIBILITY

Solution compatibility	Glucose 10%, Sodium Chloride 0.9%, Sodium Chloride 0.45%
Solution incompatibility	
IV Line compatibility	Aminophylline, Adrenaline, Caspofungin, Clonidine, Dobutamine, Gentamicin, Esmolol, Fentanyl, Furosemide (sodium chloride only), Heparin, Magnesium Sulphate, Meropenem, Midazolam, Milrinone, Morphine, Labetalol, Potassium Chloride, Sodium Bicarbonate, TPN, Vancomycin, Vecuronium
IV Line incompatibility	Digoxin, Dopamine, Noradrenaline, Omeprazole

THIS LIST IS NOT EXHAUSTIVE PLEASE CONTACT PHARMACY FOR FURTHER INFORMATION ON COMPATIBILITY WITH ANY MEDICINES NOT INCLUDED

CAUTIONS, CONTRA-INDICATIONS AND SIDE EFFECTS
See Summary of Product Characteristics and most recent edition of BNF for Children
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FURTHER INFORMATION

- Hyperglycaemia can be enhanced by the following drugs; levothyroxine, steroids, diuretics, including thiazides.
- Concentrations <1 unit/ml may cause significant line adherence
- Unopened vials must be stored in the fridge
- Once opened, vials may be used for up to 6 weeks, store below 25℃ and do not refrigerate
- Warning: do not flush through a line containing insulin once connected to the patient.

PH 7.4

LICENSED STATUS Licensed for use in all ages.

APPLICABLE POLICIES West of Scotland Neonatal Guidelines
Consult local policy if applicable

NEONATAL West of Scotland Drug Monographs
PARENTERAL ROUTE

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Administer reconstituted solutions immediately.
All vials, ampoules and infusion bags are for single use only unless otherwise stated.

Dose may vary depending on indication, age, renal function, hepatic function, and concomitant medications.
This monograph should be used in conjunction with the terms of reference document prepared by the West of Scotland Pharmacist Group. Information is correct at the time of publication and as per local practice agreement.
For further advice please contact your clinical pharmacist or pharmacy department